



Alberta Public Health Disease Management Guidelines

Diphtheria



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Communicable Disease Control Branch

Public Health Division

Alberta Health

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Revisions

Revision Date	Document Section	Description of Revision
February 2025	General	Entire guideline reviewed and updated to align with current information
	Case Definition and Diagnosis	Clarification of testing process and results
	Reporting Requirements	Updated to include lab surveillance of toxigenic <i>C. ulcerans</i> and <i>C. pseudotuberculosis</i>
	Treatment	- Removed erythromycin - Removed detailed treatment regimens

Case Definition

Confirmed Case

- A person with clinical illness^(A) **AND** any of the following:
 - confirmation of toxin-producing *Corynebacterium diphtheriae* (*C. diphtheriae*) by modified Elek testing^(B) from the nose or throat,
- OR
- epidemiologic linkage to a laboratory-confirmed case of diphtheria.

OR

- An infection at a non-respiratory anatomical site (e.g., skin, wound, conjunctiva, ear, genital mucosa) in the absence of a more likely etiology **AND** confirmation of toxin producing *C. diphtheriae* by modified Elek testing^(C) from that site.

Probable Case

- In the absence of a more likely diagnosis, clinical illness^(A) with each of the following:
 - an adherent pseudo-membrane of the nose, pharynx, tonsils, or larynx, **AND**
 - absence of laboratory confirmation, **AND**
 - lack of epidemiologic linkage to a laboratory-confirmed case of diphtheria.

OR

- Histopathologic diagnosis.

^(A) Clinical illness is characterized as an upper respiratory tract infection (nasopharyngitis, laryngitis or tonsillitis) with or without adherent nasal, tonsillar, pharyngeal and/or laryngeal membrane, plus one of the following:

- Gradually increasing stridor
- Cardiac and/or neurological involvement one to six weeks after onset
- Death, with no known cause

^(B) There are some strains of *C. diphtheriae* that have a truncated version of the diphtheria toxin gene that does not produce a functional toxin. This is evident when the genotypic (NAT) assay is positive (i.e., detects the diphtheria toxin gene), but the gold-standard phenotypic assay (modified Elek) is negative. These are referred to as “non-toxigenic, tox-gene-bearing” (NTTB) strains. The modified Elek test at the National Microbiology Laboratory (NML) is the definitive test for toxigenicity. Clinicians should call the Microbiologist on call (MOC) at Alberta Precision Laboratories (APL) to clarify any discordant results.

^(C) Due to the limitation of NAT results in identifying toxin-producing *C. diphtheriae* as noted in footnote ^(B) and the lack of evidence of toxigenic *C. diphtheriae* strains in Alberta in over a decade ⁽¹⁾, *C. diphtheriae* isolates from non-respiratory sources will **not** undergo additional NAT testing unless specifically requested by the most responsible practitioner (MRP) and/or Zone Medical Officer of Health (MOH) in consultation with the MOC at APL.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by fastest means possible (FMP).

Laboratories

All laboratories shall report all **case defining** positive laboratory results:

- by FMP to the MOH (or designate) of the zone, and
- by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Lab Surveillance ONLY:

- Isolation of other toxigenic *Corynebacterium* species (*C. ulcerans* or *C. pseudotuberculosis*) from a respiratory specimen (including the exudative membrane)

Alberta Health Services and First Nations Inuit Health Branch

- The MOH (or designate) where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and probable cases.
- The MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country case/contacts, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable), and
 - other relevant clinical / epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Diphtheria is caused by *C. diphtheriae*.^(2–4) The primary public health concern lies with those strains that carry the gene for diphtheria toxin production, referred to as “toxigenic” or “toxin-producing” strains.⁽⁵⁾

Clinical Presentation

Diphtheria is an acute bacterial disease primarily affecting the respiratory mucosa. The diphtheria toxin can cause severe respiratory illness and systemic complications. Cutaneous infections can be caused by either toxigenic or non-toxigenic strains.^(1,2) However, for both manifestations of the infection, only confirmed toxigenic^(See footnotes page 5, B,C) strains require public health action. For details on symptoms see [Table 1: Respiratory Diphtheria](#) and [Table 2: Cutaneous Diphtheria](#)

Absorption of diphtheria toxin causes systemic complications. Toxin is disseminated via blood stream from site of infection and causes damage to distant tissues and organs.^(6,7) The most common complications are myocarditis, neural toxicity and renal failure.^(2,3,8)

Before treatment was available, respiratory diphtheria had a case fatality rate of 50%; however, with treatment the rate is now 5–10%.^(2,3,6,8)

Table 1: Respiratory Diphtheria

Types	Disease Description
Nasal ^(2,7)	• Resembles a common cold with nasal discharge and a white membrane on the nasal septum. • Overall, presents as a mild infection due to poor absorption of toxin.
Pharyngeal/Tonsillar ^(2,3,7) (most common)	• Starts gradually, affecting pharynx and tonsils. • Early symptoms: sore throat, low-grade fever, anorexia and malaise. • A white, glossy pseudo-membrane appears 2-3 days later in the pharyngeal/tonsillar area; turning into a dirty gray with patches of black or green necrosis; dislodging it causes bleeding. • Severe cases may have a ‘bull neck’ due to the swelling of the anterior neck and submandibular area.
Laryngeal ^(7,8)	• May manifest as extension of pharyngeal infection or occur on its own. • Symptoms include hoarseness, fever, respiratory stridor, cyanosis, barking cough, increased restlessness and anxiety. • The membrane can cause airway obstruction, leading to coma and death.

Table 2: Cutaneous Diphtheria^(7–10)

Disease Description
• Infection is usually mild and invasive/systemic disease is rare. • Lesions begin as vesicles and develop into well-defined ulcers resembling impetigo and are usually covered by a slightly raised, dirty gray membrane. • Usually present as chronic non-healing ulcers, sometimes with other pathogens (group A streptococci and <i>Staphylococcus aureus</i>).

Diagnosis

Diagnosis is usually based on history, clinical presentation and laboratory testing. The successful confirmation of toxigenic *C. diphtheriae* depends on appropriate specimen collection. Consequently, clinicians are advised to call the MOC at APL **prior** to specimen collection for specific instructions on how to collect and submit specimens. Refer to the [Alberta Precision Laboratories Test Directory](#) for more information on specimen collection recommendations. When *C. diphtheriae* is identified in respiratory specimens, the isolate will be screened for the presence of the diphtheria toxin gene using nucleic acid testing (NAT).^(See footnote page 5, B) Only respiratory isolates that possess the toxin gene are forwarded to the National Microbiology Laboratory (NML) for the gold standard phenotypic toxin test (modified Elek test), which confirms the production of functional diphtheria toxin.^(See footnote page 5,C)

Isolation of *C. diphtheriae* from cutaneous specimens will only undergo additional NAT testing when specifically requested by the MRP and/or by zone MOH request in consultation with the MOC at APL. ^(See footnote page 5, D)

All diagnostic specimens should be collected **BEFORE** starting treatment.⁽¹¹⁾

Treatment

Type	Description of Treatment
Respiratory	Diphtheria Antitoxin - Diphtheria antitoxin (DAT) is considered the mainstay of treatment for respiratory diphtheria as it neutralizes circulating toxin and prevents progression of disease. - Treatment should begin as soon as possible based on clinical symptoms and not wait for laboratory confirmation. Collect specimens prior to starting treatment, if possible. - A person's eligibility for DAT will be determined through discussion between the treating physician, zone MOH and CMOH. Refer to Appendix 1: Diphtheria antitoxin (DAT) procurement.^(E) - For more information on use of DAT, refer to the Alberta Immunization Policy (AIP) and the product information sheet.
	Antibiotics - Antibiotic therapy with a macrolide such as azithromycin or clarithromycin is required to eradicate the organism, to stop toxin production and prevent further transmission. Antimicrobial susceptibility testing may be required. - Cases that receive treatment should have two cultures from the nose and throat collected at least 24 hours apart and at least 24 hours after completion of antibiotics.^(7,8) - If cultures are positive after the first course of antibiotics, another course is required with repeat follow-up cultures (as described in bullet above) after completion.⁽¹¹⁾ NOTE: If cultures are still positive after the additional course of antibiotics, consult with an infectious disease specialist on further antibiotic treatment recommendations.
Cutaneous	Antibiotics - DAT is not recommended for cutaneous diphtheria. - Cleanse lesion thoroughly with soap and water. - Antibiotic treatment with a macrolide such as azithromycin or clarithromycin is recommended. Antimicrobial susceptibility testing may be required - Obtain two cultures from nose, throat and skin lesions at least 24 hours apart and at least 24 hours after completion of antibiotic therapy. - If cultures are positive after first course of antibiotics, another course should be given with repeat follow-up cultures (as described in bullet above) after completion. NOTE: If cultures are still positive after an additional course of antibiotics, consult with an infectious disease specialist on further antibiotic treatment recommendations.

^(D) Clinical indications for request may include cases where *C. diphtheriae* is the predominant pathogen (in the absence of more common pathogens such as *S. aureus* or *S. pyogenes*); or when it is isolated repeatedly from subsequent specimens.

^(E) The final decision to administer DAT lies with the treating physician.

Reservoir

Humans are the only known host for *C. diphtheriae*.^(2,11)

Transmission

The most common modes of transmission are via respiratory droplets or direct contact with either respiratory secretions or exudate from infected skin lesions.^(2,3)

Incubation Period

The incubation period for respiratory diphtheria is, typically two to five days with a range of one to 10 days.^(2,3) The incubation period for cutaneous diphtheria may be longer as it is not well defined.⁽¹¹⁾

Period of Communicability

The period of communicability is variable; however, cases are usually infectious as long as virulent bacteria are present in respiratory secretions, usually two weeks or less and seldom more than four weeks without treatment.⁽²⁾ After 48 hours of appropriate antibiotic therapy, cases are no longer communicable.⁽³⁾ Without effective antibiotic therapy, asymptomatic chronic carriers colonized with *C. diphtheriae* in the nasopharynx or the skin may shed the bacteria for six months or longer.⁽²⁾

Host Susceptibility

Travelers to diphtheria endemic areas who are not adequately immunized are at a higher risk of infection.^(3,6) Diphtheria immunization protects individuals against the effects of diphtheria toxin but does not prevent them from acquiring the bacteria that cause infection. However, immunized individuals who do become infected are more likely to be asymptomatic or have milder symptoms.⁽¹¹⁾

Natural infection does not confer lifelong immunity therefore eligible individuals who have recovered from diphtheria infection should be offered diphtheria vaccine as per the [AIP](#).^(2,7)

Incidence

For information on worldwide incidence refer to the Pan American Health Organization/World Health Organization (PAHO/WHO) website: [Diphtheria](#).

In Canada, diphtheria became a nationally notifiable disease in 1924. With widespread immunization in Canada since 1930, incidence has greatly reduced and is now rare.⁽⁶⁾ For information on Canadian incidence, refer to the [Health Canada Notifiable Disease Online](#) website.

In Alberta, diphtheria became notifiable in 1985. Toxigenic respiratory diphtheria cases are very rare in Alberta; the last case occurred prior to 1995. Detection of cutaneous diphtheria is more common; however, a toxin producing infection has not been identified in Alberta for more than 10 years.⁽¹⁾

Public Health Management

Key Investigation

- Confirm that the individual meets [case definition](#).^(F)
- Obtain history of illness including date of onset of signs and symptoms.
- Determine incubation period and period of communicability.
- Determine possible transmission settings (e.g., childcare settings, homeless shelters, overcrowded housing) during the period of communicability.
- Determine diphtheria-specific immunization history:
 - number of doses,
 - date administered,
 - where the person was immunized (e.g., out of country), and
 - if not immunized, determine reason why.
- Determine possible source of infection, including:
 - recent contact with a confirmed, probable or suspect case,
 - recent travel history (during incubation period),
 - recent immigration from an area with known endemic disease,^(G)
 - members in the household with similar symptoms.
- Identify [contacts](#) and assess immunization history.
 - The risk of transmission is directly related to the closeness, duration of contact, and intensity of the exposure.
 - Notify MOC at APL as soon as diagnosis is suspected to ensure appropriate clinical specimen(s) are collected in appropriate media prior to commencing treatment.

Table 3: Contacts Definition

Contacts
Individuals who, in the last 10 days^(H) were: • living in the same household or share sleeping arrangements, • sexual contacts, • in group living settings and shared a bathroom with the case or had close interactions with the case (for example, dormitories, shelters, detentions centers, group homes, settlement houses), • other individuals who had direct contamination of the nose or mouth with oral and/or nasal secretions of the case (e.g., kissing, shared utensils), • children and staff in childcare and preschool facilities, • in contact with nasopharyngeal secretions of the case including, health care workers (HCW)^(I) without the use of adequate personal protective equipment (PPE) and infection prevention and control (IPC) practices (i.e., determined to have breaches in PPE), • laboratory workers who handled specimens (i.e., throat swabs, nasopharyngeal swabs/aspirates, urine) without the use of adequate personal protective equipment (PPE) and infection prevention and control (IPC) practices (i.e., determined to have breaches in PPE), • regular visitors to the home (e.g., grandparents, housekeeper, tutor).

^(F) Cutaneous diphtheria detection by NAT should not prompt public health action; confirmatory Elek testing by NML is necessary to identify toxin-producing strains that require reporting and public health investigation.

^(G) See [Diphtheria - WHO for global incidence of diphtheria](#) however, recent local epidemiology may not yet be reflected in information presented. Public Health Notices - Canada may provide recent information for diphtheria outbreaks occurring in Canada.

^(H) This refers to the 10 days prior to the date the specimen was sent for ELEK testing, regardless of when it was originally collected. This timeframe reflects a high index of suspicion for toxigenic *C. diphtheriae* even though toxin testing results are still pending

^(I) Refer to the [Alberta Immunization Policy \(AIP\)](#) for [HCW eligibility](#).

Management of a Case

- Provide information about disease transmission and measures to minimize transmission.
- Hospitalized cases should be managed in accordance with hospital infection prevention and control guidelines; additional guidance provided in [Table 4: Case Management Precautions](#)
- Ask cases to pay strict attention to personal hygiene by:
 - covering mouth and nose with tissue when coughing,
 - placing all contaminated tissues directly into garbage,
 - washing hands with soap and water every time there is contact with respiratory secretions or infected wounds, and
 - keeping all infected wounds covered.
- Exclusion: Refer to [Table 5: Case Exclusion Criteria](#).

Table 4: Case Management Precautions

Case	Precautions
Respiratory	Hospitalized case: • Should be managed in accordance with hospital infection prevention and control guidelines
	Non-hospitalized case^(A): • Contact precautions apply until two cultures from both nose and throat, taken at least 24 hours apart and at least 24 hours after cessation antimicrobial therapy, are negative. • Individuals with non-severe disease can be treated and followed by a community physician with support from public health professionals. Consultation with an infectious disease specialist may be required, to determine appropriate course of treatment and follow-up.
Cutaneous	Hospitalized case: • Should be managed in accordance with hospital infection prevention and control guidelines
	Non-hospitalized case^(A): • Individuals are most commonly treated and followed by a community physician with support from public health professionals. Consultation with an infectious disease specialist may be required to determine appropriate course of treatment and follow-up. • Recommend minimal contact with others until two cultures from both nose, throat and skin lesions, taken at least 24 hours apart and at least 24 hours after cessation of antimicrobial therapy, are negative. • Articles in contact with infected individual and articles soiled by discharges of the case should be washed as per normal practices.

^(A) Examples of non-hospitalized: community/home settings, long-term care/assisted living settings, shelter.

Table 5: Case Exclusion

Case Exclusion
• The MOH shall exclude cases from workplace, school or childcare settings until antibiotic therapy is completed and two cultures from nose, throat and/or lesions collected at least 24 hours apart and at least 24 hours after cessation of antimicrobial therapy are negative (if skin lesion/wound has healed, swab skin where lesion/wound is located).

Management of Contacts

- Confirm exposure history and the time since last exposure to the case.
- Determine immunization history (i.e., type of vaccine, number of doses and date of administration), offer immunization as appropriate as per [AIP](#).
- [Contacts](#) exposed to the same source (e.g., case, location) should be educated about the disease including prevention, and advised to connect with their healthcare provider or call HealthLink at 811 if symptoms occur.
- Provide information about diphtheria disease including signs and symptoms.
- All contacts should have a single swab for culture taken from each of the nose, the throat and skin lesions (where present).
- Collect swabs **prior** to initiating antibiotic prophylaxis. See post-exposure prophylaxis section for more details.
- Refer symptomatic contacts for assessment.
- Any asymptomatic contact identified as having a positive swab for toxigenic diphtheria should be treated as a carrier (see [Appendix 2: Management of Diphtheria Contacts and Carriers](#).)
- Advise asymptomatic contacts to monitor closely for symptoms for at least 10 days after their last exposure to the infected person and to connect with their healthcare provider or call HealthLink at 811 if symptoms occur. They should also notify public health if they develop symptoms.
- Contacts not up-to-date for diphtheria immunization should be offered age-appropriate diphtheria containing vaccine according to the current [AIP](#).
- Refer to Table 6 for contact exclusion criteria.

Table 6: Contact Exclusion

Contact Exclusion
• The MOH shall exclude contacts from the following pending first set of negative specimen results: - contact with children, - occupations involving the care of the sick and dependent, - occupations involving the handling of food, and - school. NOTE: For more information refer to Appendix 2: Management of Contacts and Carrier Algorithm.

Post-Exposure Prophylaxis (PEP) of Contacts

- PEP should be offered to all contacts after swabs from the nose, throat and skin lesions (where present) have been taken, regardless of their immunization status.
 - Contacts are given PEP to treat incubating disease and eliminate carriage, thereby reducing risk of transmission to other susceptible contacts.
- Antibiotic treatment recommended for PEP is a macrolide such as azithromycin or clarithromycin.
- There is no acceptable clinical evidence for prophylactic use of DAT for contacts.
- Eligible contacts should receive diphtheria-containing vaccine as per the *AIP* to ensure they are up-to-date for diphtheria immunization.

Management of Carriers

- A carrier is defined as a person who harbors and may transmit *C. diphtheriae* but who has no symptoms. Carriers can include those with otitis media or nasal infections and asymptomatic pharyngeal infection due to toxigenic *C. diphtheriae*.
- Carriers are **NOT** reportable to Alberta Health.
- The MOH **shall exclude** carriers from the following until completion of antibiotic therapy and **two** cultures from nose, throat and/or lesions (when present) are collected at least 24 hours apart and at least 24 hours after cessation of antimicrobial therapy are negative:
 - contact with children,
 - occupations involving the care of the sick and dependent,
 - occupations involving the handling of food, and
 - school.

NOTE: For more information, refer to [Appendix 2: Management of Contacts and Carriers Algorithm](#).

- If repeat cultures are positive, recommend an additional course of antibiotics and continue with exclusion.
- **NOTE:** If cultures are positive after additional course of antibiotics, consult with an infectious disease specialist on further antibiotic treatment recommendations.
- Carriers should be instructed to pay strict attention to personal hygiene by:
 - covering mouth and nose with tissue when coughing,
 - placing all contaminated tissues directly into garbage,
 - washing hands with soap and water every time there is contact with respiratory secretions or infected wounds,
 - cleaning wounds and skin lesions vigorously with soap and water, and
 - keeping all infected wounds covered.

Treatment of Carriers

- Carriers should be given antibiotic treatment regardless of immunization status.
- Recommended antibiotics: a macrolide such as azithromycin or clarithromycin.
- DAT has no proven role in the treatment of diphtheria carriers.
- If not fully immunized, all carriers should be offered age-appropriate diphtheria containing vaccine according to the current [AIP](#) with measures taken to ensure completion of series.

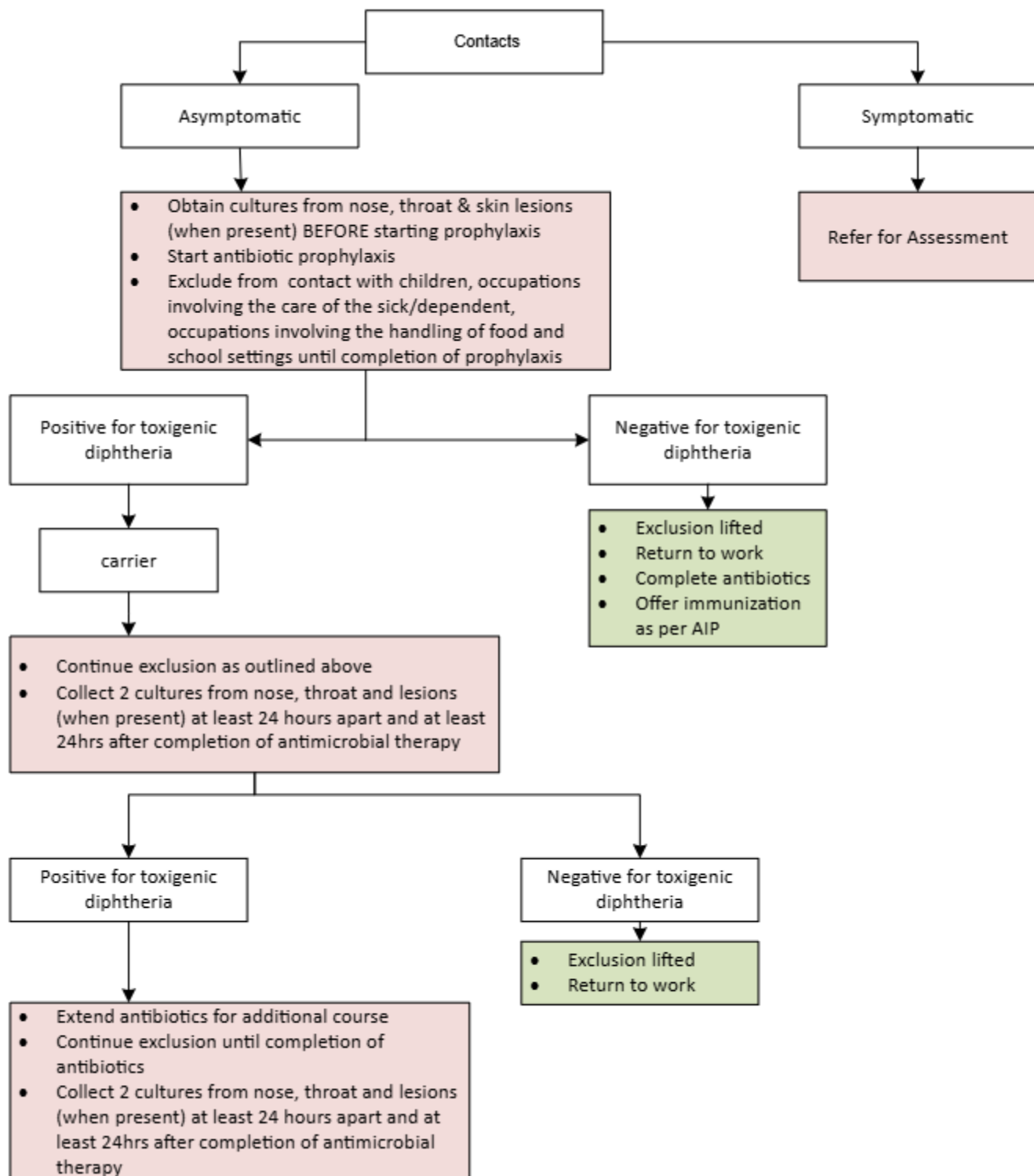
Preventive Measures

- Educate the public on the risks of diphtheria and how to prevent transmission including:
 - practicing good hand hygiene and respiratory etiquette,
 - avoiding sharing drinks or any other items used on the nose or mouth, and
 - cleaning frequently touched household surfaces.
 - keeping immunizations up to date, Including relevant travel vaccines prior to departure.
 - refer to the [AIP](#) for current diphtheria vaccine recommendations.
- Remind individuals planning on taking a trip to seek travel advice prior to leaving, especially if going to a location with known diphtheria cases/outbreak(s). Refer to the Government of Canada's [Travel Advice and Advisories by destination website](#).

Appendix 1: DAT Procurement

- Currently there is no licensed product made in Canada. DAT is made available from Health Canada's Special Access Program (SAP).
- The MOH (or designate) must complete an [SAP "Form C"](#) (patient follow-up form) and send it back to the CMOH: this form will accompany all shipments of DAT.
- The CMOH will communicate with the Immunization Team at Alberta Health during work hours or directly with the Provincial Vaccine Depot (VPD) on call after hours to facilitate access and shipment under strict cold-chain management from the Provincial Vaccine Depot (PVD) to the facility pharmacy by the most rapid mode of transport available.
- The CMOH will contact Health Canada's SAP for urgent delivery of more vials if required.
- **Please return if the product is not used.** If the DAT is not used, the product can be shipped (maintaining cold-chain) back to the PVD.

Appendix 2: Management of Diphtheria Contacts and Carriers



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